UBICOMP ISWC2025

14-16 Oct. 2025
Helsinki, Finland

Summary and outlook

Everyone

OPEN AR CLOUD

The future: Spatial Internet?

Digital contents, links, apps mixed with the physical world in both directions

What infrastructure and services are required?

Implications on privacy and on our lives?



Summary and future work

We are creating open-source building blocks for an open and interoperable spatial Web
https://github.com/OpenArCloud, MIT license

Universities are great targets for deployment:

- Tool/platform for location-based AR research, student developers, hackathons
- Campus environment, live content sources

We are seeking researcher + developer collaborators for

- (Visual) positioning services
- Spatial discovery (IoT devices, complex scenes, etc.)
- Local-first software: share, process, store everything locally, encapsulate data (Named Data Networking?, etc.)
- Reality modeling (semantics, level of detail, privacy?, etc.)

Future research

Distributed infrastructure

- Edge compute vs. device compute vs. hybrid split
- Map and content distribution, efficient search
- Dynamic resource allocation, device-to-device sharing

Localization

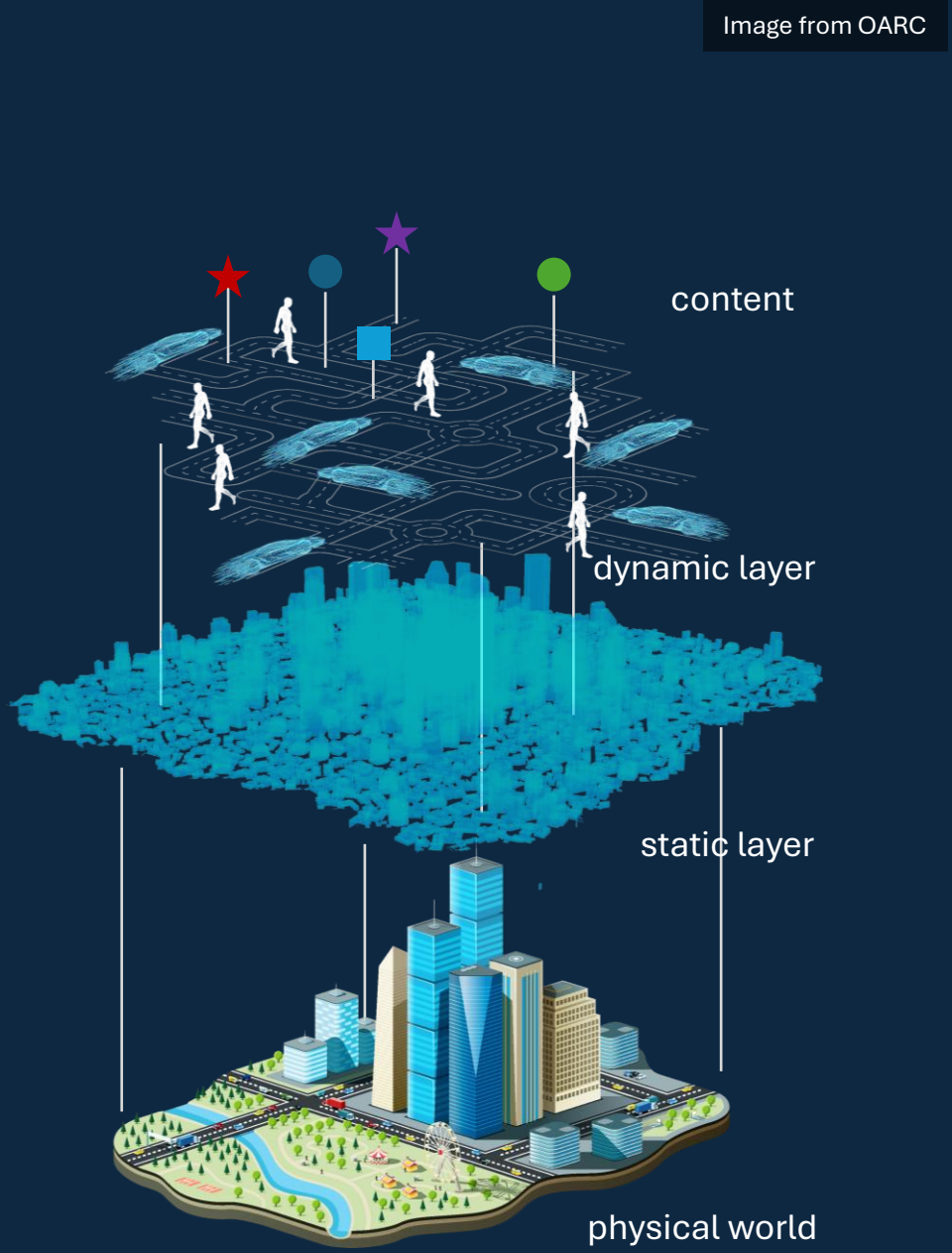
- Radio (+ GPS) + vision (~1cm, ~1deg)
- Queries as updates for a digital city model
- Wireless network planning

Reality modeling

- Environment capture, 3D reconstruction, Semantic understanding
- Machine-readable vs. human-readable representations

Rendering

- Situated visualization: 4D data in spatial context



Future research

Applications

- Gamification of mapping
- Tourism, navigation, art, smart city services
- Low-latency multi-user applications

Privacy and security

- Location anonymization, history
- Map ownership, map anonymization
- Content creation, sharing, access control
- Blockchained maps and content
- GeoPose attacks, AR spam

Devices, etc.



Information systems start small and grow. They also start isolated and then merge. A new system must allow existing systems to be linked together without requiring any central control or coordination. /Tim Berners-Lee/

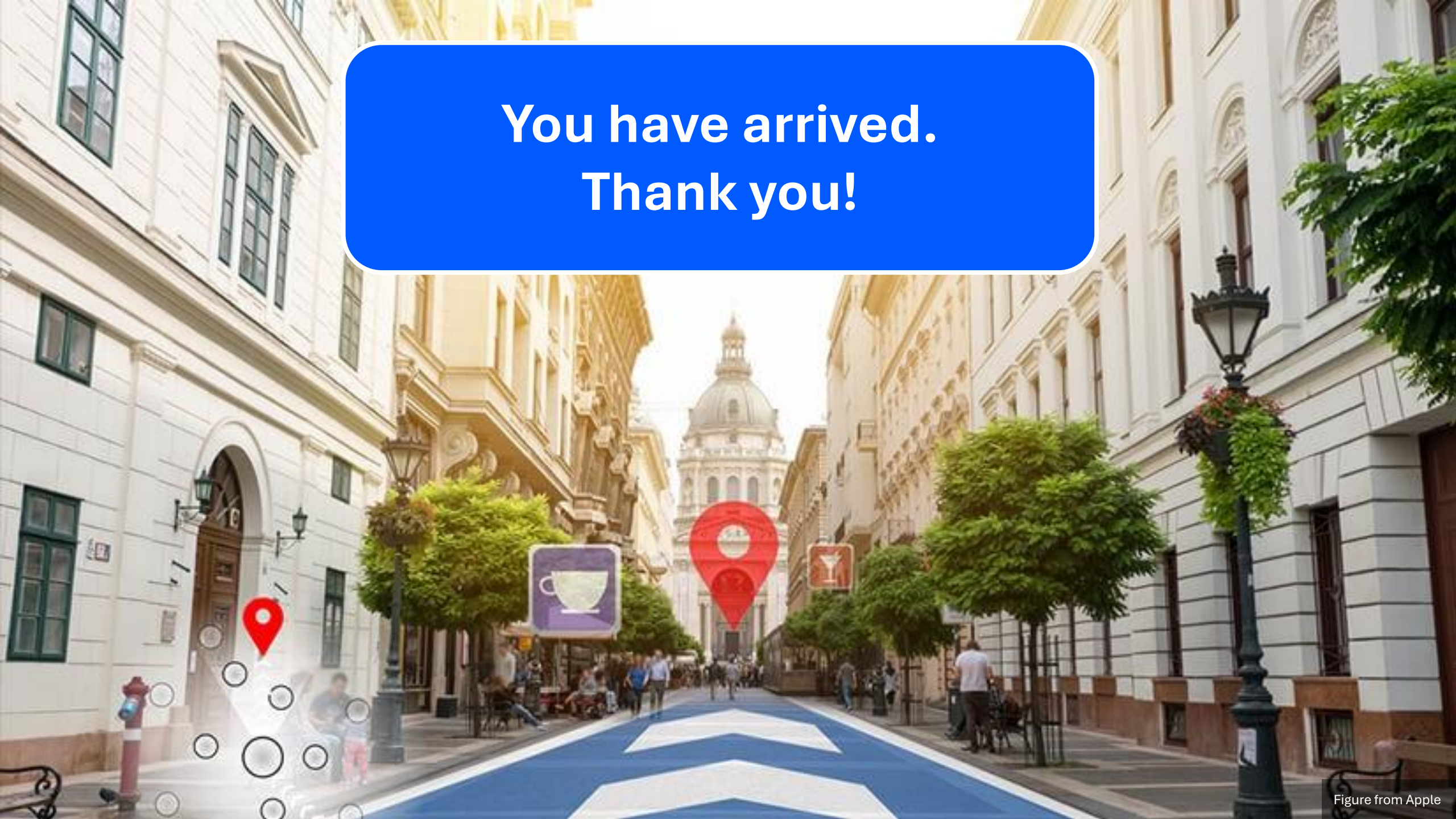


Figure from Apple